

Python OOP and Function Assessment (Case Study Based)

Section 1: Functions and OOP Concepts

1. 1. Case Study: Grocery Store Billing
A grocery store wants to automate billing. Create a function to calculate the total bill amount for a list of items with their prices and quantities.
2. 2. Case Study: School Report Generator
A school wants to generate student reports using a function. Write a function that takes student name, marks in 3 subjects and returns the average marks and grade.
3. 3. Case Study: Temperature Converter
Create a Python program using a function that converts temperature from Celsius to Fahrenheit and vice versa based on user choice.
4. 4. Case Study: Employee Management System
An organization wants to create a simple employee management system. Design a class Employee with attributes like name, ID, and salary. Add a method to display the details.
5. 5. Case Study: Bank Account Simulation
Write a class BankAccount that allows a user to deposit, withdraw, and check balance. Use functions for each operation.
6. 6. Case Study: Movie Ticket Booking System
Design a class MovieTicket with attributes like movie name, seat number, and price. Include a method to calculate total price for multiple tickets.
7. 7. Case Study: Library Book Management
A library wants to keep track of books. Create a class Book with attributes like title, author, and availability. Include methods to borrow and return a book.
8. 8. Case Study: Car Rental System
Write a class Car with attributes like brand, model, and daily rent. Create a function to calculate rent for a given number of days.
9. 9. Case Study: Online Course Progress Tracker
Create a class Course with attributes course name, total modules, and completed modules. Write a method to display the percentage of course completion.
10. 10. Case Study: Restaurant Order System
Design a function to accept a list of food items ordered and their prices, then return the total bill amount and number of items ordered.

Section 2: OOP Core Concepts (Abstraction, Encapsulation, Inheritance, Polymorphism)

11. 1. Abstraction – Case Study: ATM Machine

An ATM machine provides only limited operations to the user such as withdraw, check balance, and deposit. Design a class where internal processing details like verifying the PIN, maintaining logs, or updating central server are hidden from the user. The user should only interact with essential methods.

12. 2. Encapsulation – Case Study: Student Record System

A school system needs to protect student data like marks and personal info. Design a class Student that stores name and marks, ensuring marks cannot be directly accessed or changed from outside the class. Include setter and getter methods to modify and view marks.

13. 3. Inheritance – Case Study: Vehicle Types

A vehicle rental company manages various types of vehicles: cars, bikes, and trucks. Design a base class Vehicle with common properties like brand and model, and then create child classes Car, Bike, and Truck inheriting from Vehicle, each with specific features or methods.

14. 4. Polymorphism – Case Study: Payment Gateway

An e-commerce platform allows multiple types of payments – credit card, debit card, and UPI. Create a base class Payment with a method make_payment(). Each payment method should override this method to show a different payment process.